# Homework 8

EE 999 — Introduction to Control Systems

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*Due Wednesday, November 11, 2026*

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## Problem A — PI and PD Control

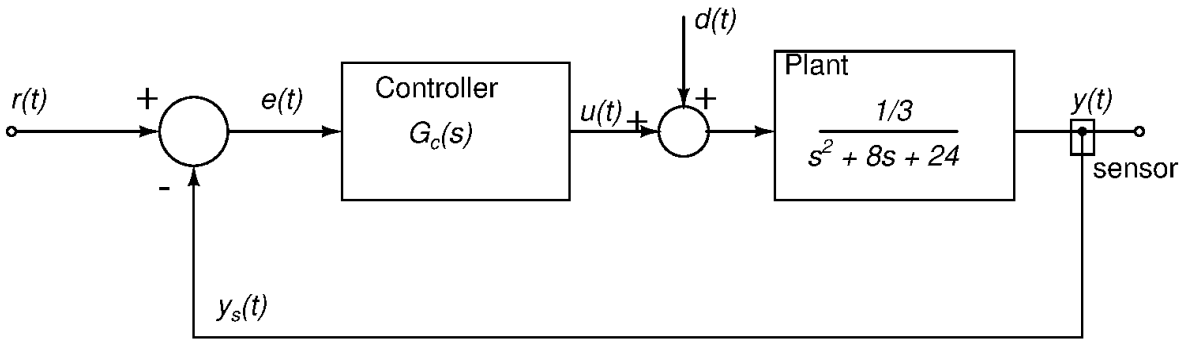



Figure 1: A system with PI-type control.

## Problem A.1 — PI Control

*Use* $G_c(s) = \dfrac{k_p s + k_i}{s}$ *with* $k_p = 10$, $k_i = 200$.

## Problem A.1.a

*Determine the transfer functions:*

**i)** $T_{ry}(s)$**, the input-output transfer function**

$$G_p(s) = \frac{1/3}{s^2 + 8s + 24} \tag{1}$$

$$G_c(s) = \frac{k_p s + k_i}{s} \tag{2}$$

$$Y(s) = \frac{G_p G_c}{1 + G_p G_c} R(s) \tag{3}$$

$$Y(s) = \frac{\dfrac{1/3}{s^2 + 8s + 24} \cdot \dfrac{10s + 200}{s}}{1 + \dfrac{1/3}{s^2 + 8s + 24} \cdot \dfrac{10s + 200}{s}} R(s) \tag{4}$$

$$Y(s) = \frac{\frac{1}{3}(10s + 200)}{s(s^2 + 8s + 24) + \frac{1}{3}(10s + 200)} R(s) \tag{5}$$

$$Y(s) = \frac{10s + 200}{3s(s^2 + 8s + 24) + 10s + 200} R(s) \tag{6}$$

$$Y(s) = \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200} R(s) \tag{7}$$

$$T_{ry}(s) = \frac{Y(s)}{R(s)} = \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200} \tag{8}$$

**ii)** $T_{re}(s)$**, the error transfer function**

$$E(s) = R(s) - Y(s) \tag{9}$$
$$E(s) = R(s) - T_{ry}(s)R(s) \tag{10}$$
$$E(s) = \left(1 - \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200}\right) R(s) \tag{11}$$
$$E(s) = \left(\frac{3s^3 + 24s^2 + 82s + 200}{3s^3 + 24s^2 + 82s + 200} - \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200}\right) R(s) \tag{12}$$
$$E(s) = \frac{3s^3 + 24s^2 + 72s}{3s^3 + 24s^2 + 82s + 200} R(s) \tag{13}$$
$$T_{re}(s) = \frac{E(s)}{R(s)} = \frac{3s^3 + 24s^2 + 72s}{3s^3 + 24s^2 + 82s + 200} \tag{14}$$

**iii) $T_{dy}(s)$, the disturbance transfer function**

$$Y(s) = G_p \left(D(s) + U(s)\right) \tag{15}$$
$$Y(s) = G_p \left(D(s) + G_c(s)E(s)\right) \tag{16}$$
$$Y(s) = G_p \left(D(s) - G_c(s)Y(s)\right) \tag{17}$$
$$Y(s) = G_p D(s) - G_p G_c(s) Y(s) \tag{18}$$
$$Y(s) + G_p G_c(s) Y(s) = G_p D(s) \tag{19}$$
$$Y(s) \left(1 + G_p G_c(s)\right) = G_p D(s) \tag{20}$$
$$Y(s) = \frac{G_p}{1 + G_p G_c(s)} D(s) \tag{21}$$
$$Y(s) = \frac{\dfrac{1/3}{s^2 + 8s + 24}}{1 + \dfrac{1/3}{s^2 + 8s + 24} \cdot \dfrac{10s + 200}{s}} D(s) \tag{22}$$
$$Y(s) = \frac{\frac{1}{3}s}{s(s^2 + 8s + 24) + \frac{1}{3}(10s + 200)} D(s) \tag{23}$$
$$Y(s) = \frac{s}{3s(s^2 + 8s + 24) + 10s + 200} D(s) \tag{24}$$
$$Y(s) = \frac{s}{3s^3 + 24s^2 + 72s + 10s + 200} D(s) \tag{25}$$
$$Y(s) = \frac{s}{3s^3 + 24s^2 + 82s + 200} D(s) \tag{26}$$
$$T_{dy}(s) = \frac{Y(s)}{D(s)} = \frac{s}{3s^3 + 24s^2 + 82s + 200} \tag{27}$$

## Problem A.1.b

*Draw a pole-zero diagram of $T_{ry}(s)$, label the dominant mode.*

**Code:** `hw_08_A1b.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

num = [10 200];
den = [3 24 82 200];
sys = tf(num, den);

pzmap(sys);

sys_poles = pole(sys)
sys_zeros = zero(sys)
```

**Output:** `output/hw_08_A1b.txt`

```
sys_poles =

  -5.2097 +      0i
  -1.3952 + 3.2940i
  -1.3952 - 3.2940i

sys_zeros = -20
```

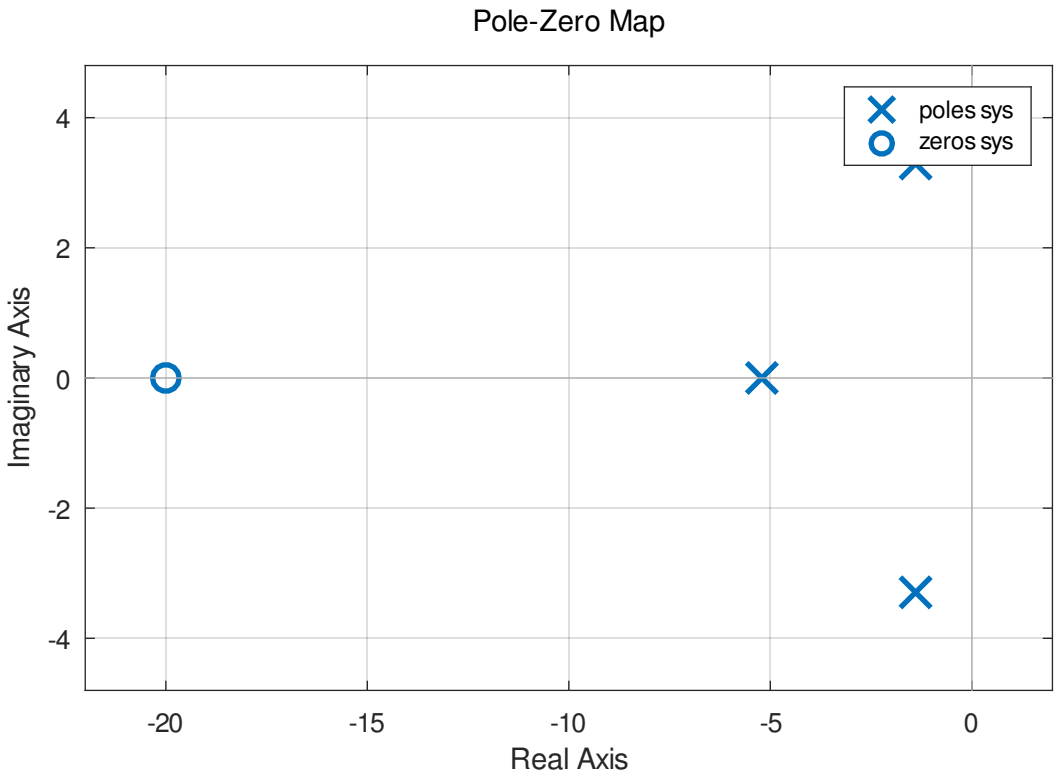



Figure 2: Pole-zero diagram of $T_{ry}(s)$.

The dominant poles are at $s = -1.3952 \pm 3.2940j$.

## Problem A.2

*Based on the dominant mode pole locations, determine*

$$p_{1,2} = -\alpha \pm j\omega \tag{28}$$
$$= -1.3952 \pm 3.2940j \tag{29}$$

$$\omega_n = \sqrt{\alpha^2 + \omega^2} \tag{30}$$
$$= \sqrt{1.3952^2 + 3.2940^2} \tag{31}$$
$$= 3.5773 \text{ rad/s} \tag{32}$$

**i) $t_p$, the peak time**

$$t_p = \frac{\pi}{\omega} \tag{33}$$
$$= \frac{\pi}{3.2940} \tag{34}$$
$$= 0.9537 \text{ s} \tag{35}$$

**ii) $\rho$, the damping**

$$\rho = \frac{\alpha}{\omega_n} \tag{36}$$
$$= \frac{1.3952}{3.5773} \tag{37}$$
$$= 0.3900 \tag{38}$$

**iii) $M_p$, the percent overshoot**

$$M_p = 100\% \times e^{-\pi\alpha/\omega} \tag{39}$$
$$= 100\% \times e^{-\pi(1.3952)/3.2940} \tag{40}$$
$$= 26.43\% \tag{41}$$

**iv) $t_r$, the rise time**

$$t_r \approx \frac{1.8}{\omega_n} \tag{42}$$
$$\approx \frac{1.8}{3.5773} \tag{43}$$
$$\approx 0.5032 \text{ s} \tag{44}$$

**v) $t_s$, the settling time**

$$t_s \approx \frac{4}{\alpha} \tag{45}$$
$$\approx \frac{4}{1.3952} \tag{46}$$
$$\approx 2.8670 \text{ s} \tag{47}$$

## Problem A.3

*Analytically determine $e_{ss}$, the steady state error, by determining the system type and $K_{Bode}$.*

$$G_L(s) = G_c(s)\, G_p(s)\, H_y(s) \tag{48}$$
$$= \frac{10s + 200}{s} \cdot \frac{1/3}{s^2 + 8s + 24} \cdot 1 \tag{49}$$
$$= \frac{\frac{1}{3}(10s + 200)}{s(s^2 + 8s + 24)} \tag{50}$$

$G_L(s)$ has one pole at the origin, so the system is Type I. A Type I system has no steady-state error for a step input, $e_{ss} = 0$.

## Problem A.4

*Setup $T_{ry}(s)$ in Matlab, run and plot the (true) step response. Determine these values for the complete system*

$$T_{ry}(s) = \frac{10s + 200}{3s^3 + 24s^2 + 82s + 200} \tag{8}$$

**Code:** hw_08_A4.m

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

% T_ry(s) = (10s + 200) / (3s^3 + 24s^2 + 82s + 200)
num = [10 200];
den = [3 24 82 200];
sys = tf(num, den);

info = stepinfo(sys)

damp(sys)

step(sys);
grid on;
title('Step response of T_{ry}(s)');
```

**Output:** output/hw_08_A4.txt

```
info =

  scalar structure containing the fields:

    RaiseTime = [](0x0)
    RiseTime = 0.4936
    TransientTime = 2.4754
    SettlingTime = 2.4754
    SettlingMin = 0.9000
    SettlingMax = 1.2038
    Overshoot = 20.376
    Undershoot = 0
    Peak = 1.2038
    PeakTime = 1.1429
    StaticGain = 1
    InitialJump = 0

   Pole                  Damping     Frequency       Time Constant
                                     (rad/seconds)   (seconds)

   -1.40e+00+3.29e+00i   3.90e-01    3.58e+00        7.17e-01
   -1.40e+00-3.29e+00i   3.90e-01    3.58e+00        7.17e-01
   -5.21e+00+0.00e+00i   1.00e+00    5.21e+00        1.92e-01
```

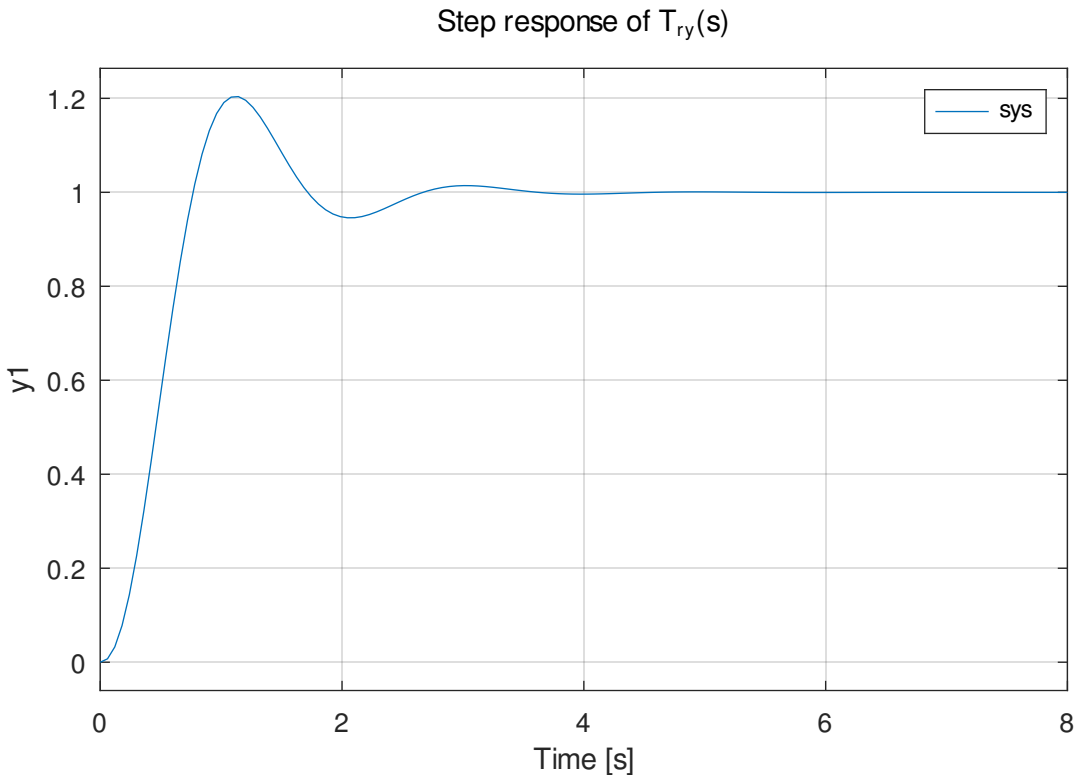



Figure 3: Step response of $T_{ry}(s)$.

**i) $t_p$, the peak time**

$$t_p = 1.1429 \text{ s} \tag{51}$$

**ii) $\rho$, the damping**

$$\rho = 0.3900 \tag{52}$$

**iii) $M_p$, the percent overshoot**

$$M_p = 20.38\% \tag{53}$$

**iv) $t_r$, the rise time**

$$t_r = 0.4936 \text{ s} \tag{54}$$

**v) $t_s$, the settling time**

$$t_s = 2.4754 \text{ s} \tag{55}$$

**vi) $e_{ss}$, the steady state error**

$$e_{ss} = 0 \tag{56}$$

## Problem A.5

*(Essay question) What is the value of the methods of steps A.1.b–A.2, relative to the method of step A.4?*

*write the answer here*

## Problem A.6

*Based on parts A.1.b–A.4, which measures based on the dominant modes came out most accurate? Which measures based on the dominant modes (part 3) came out least accurate?*

*write the answer here*

## **Problem B** — PD Control

*Use $G_c(s) = k_d s + k_p$ with $k_p = 300$, and $k_d$ to be determined.*

## **Problem B.1** — Experiment with $k_d$

### Problem B.1.a

*Derive $T_{ry}(s)$, the closed loop transfer function with PD control, and with $k_p$ and $k_d$ as variables. Set $k_p = 300$ and determine the pole locations for several values of $k_d$ on the range from 0.5 to 10.*

*write the answer here*

### Problem B.1.b

*Hand sketch a plot of the closed-loop pole locations as a function of $k_d$.*

*write the answer here*

### Problem B.1.c

*Select a value of $k_d$ to give $\rho = 0.4$, be rather accurate, e.g., $0.39 < \rho < 0.41$.*

*write the answer here*

Using the value of $k_d$ chosen to give $\rho = 0.4$:

## Problem B.2

*What is the dominant mode of this system? (Same as asking "where are the dominant pole(s)")*

*write the answer here*

## Problem B.3

*Analyze your system (analytically, based on the dominant mode) to determine:*

**i) $t_p$, the peak time**

*write the answer here*

**ii) $\rho$, the damping**

*write the answer here*

**iii) $M_p$, the percent overshoot**

*write the answer here*

**iv) $t_r$, the rise time**

*write the answer here*

**v) $t_s$, the settling time**

*write the answer here*

## Problem B.4

*Analytically determine $e_{ss}$, the steady state error, by determining the system type and $K_{Bode}$.*

*write the answer here*

## Problem B.5

*Analyze the step response of your system (step response, full system, you can use the* `step` *and* `stepinfo` *commands) to determine:*

**i) $t_p$, the peak time**

*write the answer here*

**ii) $\rho$, the damping**

*write the answer here*

**iii) $M_p$, the percent overshoot**

*write the answer here*

**iv) $t_r$, the rise time**

*write the answer here*

**v) $t_s$, the settling time**

*write the answer here*

**vi) $e_{ss}$, the steady state error (not indicated by `stepinfo()`)**

*write the answer here*

## Problem B.6

*Which of the values based on dominant mode are closest to their analysis values? Which are farthest?*

*write the answer here*

## **Problem C** — Compare the PI and PD controllers

*Consider these parameters; the desirable characteristic is indicated.*

| Controller performance measure | Desirable characteristic |
|---|---|
| i) $t_p$, the peak time | Fast |
| ii) $\rho$, the damping | Near one |
| iii) $M_p$, the percent overshoot | Near zero |
| iv) $t_r$, the rise time | Fast |
| v) $t_s$, the settling time | Fast |
| vi) $e_{ss}$, the steady-state error | Small |

## **Problem C.1**

*What characteristic is most enhanced by the PI control system?*

*write the answer here*

## **Problem C.2**

*What characteristic is most enhanced by the PD control system (tuned for $\rho = 0.4$)?*

*write the answer here*